Week 15

MICROECONOMIC THEORY
ECON 323 502/503

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Review

- **Cournot duopoly model** is a model in which two firms **simultaneously** choose **the quantity** to produce.
- The Nash equilibrium in a Cournot duopoly game is the **Cournot equilibrium**.

- **Bertrand duopoly model** is a model in which two firms **simultaneously** choose **prices** to post.
- The Nash equilibrium in a Bertrand duopoly game is the **Bertrand equilibrium**.

Stackelberg equilibrium

Stackelberg duopoly model is a model where two firms **sequentially** choose the **quantity** to produce.

The **subgame perfect Nash equilibrium** of the Stackelberg duopoly model is called **Stackelberg equilibrium**.

A Mini Example on Cournot equilibrium

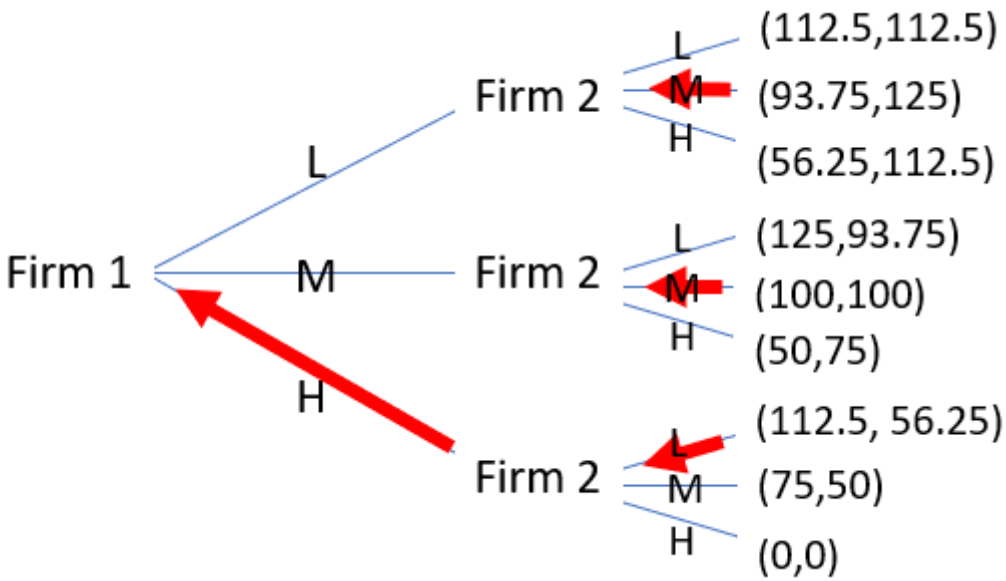
		Firm 2		
		L	M	H
Firm 1	L	112.5, 112.5	93.75, 125	56.25 112.5
	M	125, 93.75	100, 100	50, 75
	H	112.5, 56.25	75, 50	0, 0

If we view this game as a Cournot model, i.e., two firms simultaneously choose the output level to produce, then (M, M) is the Cournot equilibrium.

A Mini Example on Stackelberg equilibrium

		Firm 2		
		L	M	H
Firm 1	L	112.5, 112.5	93.75, 125	56.25 112.5
	M	125, 93.75	100, 100	50, 75
	H	112.5, 56.25	75, 50	0, 0

Suppose Firm 1 moves first, and Firm 2 moves after observing Firm 1's output choice. What are the subgame perfect Nash equilibrium payoffs in the sequential game?



The Stackelberg equilibrium:

- Firm 2: if Firm 1 plays L, I will play M; if Firm 1 plays M, I will play M; if Firm 1 plays H, I will play L.
- Firm 1: H.

Stackelberg Equilibrium Example

Suppose market demand is given by $Q(p)=1000-2p$. Cost functions of the two firms in the duopoly market are given by $C_1(q_1)=200q_1$ and $C_2(q_2)=200q_2$. Assume that Firm 1 is the leader of the game and Firm 2 is the follower. What is the Stackelberg equilibrium?

- How to solve this problem?
- Use backward induction!

Stackelberg Equilibrium Example: Solving Follower's Problem

Q(p)=1000-2p. Leader $C_1(q_1)=200q_1$. Follower $C_2(q_2)=200q_2$.

Step 1(1): Derive the inverse demand function

Step 1(2): $\boldsymbol{R_2}$

Step 1(3) : Compute $\mathbf{MR_2}$ (take derivative of $\boldsymbol{R_2}$ respect to q_2), $\mathbf{MC_2}$

Step 2: Set $\boldsymbol{MR_2 = MC_2}$

Stackelberg Equilibrium Example: Solving Follower's Problem

Q(p)=1000-2p. Leader $C_1(q_1)=200q_1$. Follower $C_2(q_2)=200q_2$.

Step 1(1): Derive the inverse demand function

- $p(q) = 500 - 0.5q$, where $q = q_1 + q_2$.

Step 1(2): $\boldsymbol{R_2}$

- $R_2 = p(q_1 + q_2)q_2 = (500 - 0.5q_1 - 0.5q_2)q_2 = 500q_2 - 0.5q_1q_2 - 0.5q_2^2$.

Step 1(3) : Compute $\mathbf{MR_2}$ (take derivative of $\boldsymbol{R_2}$ respect to q_2), $\mathbf{MC_2}$

- $\mathrm{MR}_2 = 500 - 0.5q_1 - q_2$. $\mathrm{MC}_2 = 200$.

Step 2: Set $\boldsymbol{MR_2 = MC_2}$

- $500 - 0.5q_1 - q_2 = 200$, isolate q_2: $q_2 = 300 - 0.5q_1$ (2)
- **This is the Reaction curve of Firm 2, i.e., the optimal output level of Firm 2 given Firm 1's output level.**
- Can write it as $q_2(q_1) = 300 - 0.5q_1$, i.e., q_2 is a function of q_1. In fact, higher output q_1 makes the follower produce less!

Stackelberg Equilibrium Example: Leader's Problem

Q(p)=1000-2p. Leader $C_1(q_1)=200q_1$. Follower $C_2(q_2)=200q_2$.

Step 1(1): Derive the inverse demand function

Step 1(2): $\boldsymbol{R_1}$

Step 1(3) : Compute $\mathbf{MR_1}$ (take derivative of $\boldsymbol{R_1}$ respect to q_1), $\mathbf{MC_1}$

Step 2: Set $\boldsymbol{MR_1 = MC_1}$

Stackelberg Equilibrium Example: Leader's Problem 2

Q(p)=1000-2p. Leader $C_1(q_1)=200q_1$. Follower $C_2(q_2)=200q_2$.

Step 1(1): Derive the inverse demand function

- Total output $q = q_1 + q_2(q_1) = q_1 + (300 - 0.5q_1) = 300 + 0.5q_1$,
- $p(q) = 500 - 0.5q = 500 - 0.5(300 + 0.5q_1) = 350 - 0.25q_1$

Step 1(2): $\boldsymbol{R_1}$

- $R_1 = p(q)q_1 = (350 - 0.25q_1)q_1 = 350q_1 - 0.25q_1^2$

Step 1(3) : Compute $\mathbf{MR_1}$ (take derivative of $\boldsymbol{R_1}$ respect to q_1), $\mathbf{MC_1}$

- $\mathrm{MR}_1 = 350 - 0.5q_1$. $\mathrm{MC}_1 = 200$.

Step 2: Set $\boldsymbol{MR_1 = MC_1}$

- $350 - 0.5q_1 = 200$, isolate q_1: $q_1 = 300$ **This is Firm 1's output level.**
- $q_2(q_1) = 300 - 0.5q_1 = 150$ **This is Firm 2's output level.**

Stackelberg Equilibrium Example: Solving Equilibrium

The Stackelberg equilibrium involves the following **strategy profile**:

- Firm 2 (follower) responds by playing $q_2(q_1) = 300 - 0.5q_1$ for each possible q_1.
- Firm 1 (leader) produces 300 units.

The outcome involves Firm 1 and Firm 2 producing (300, 150).

Equilibrium price level:

Profit level for Firm 1:

Profit level for Firm 2:

There is a first-mover advantage!

Stackelberg Equilibrium Example: Solution

The Stackelberg equilibrium involves the following **strategy profile**:

- Firm 2 (follower) responds by playing $q_2(q_1) = 300 - 0.5q_1$ for each possible q_1.
- Firm 1 (leader) produces 300 units.

The outcome involves Firm 1 and Firm 2 producing (300, 150).

Equilibrium price level: $p(q) = 500 - 0.5q = 500 - 0.5 * 450 = 275$

Profit level for Firm 1: $p(q)q_1 - C_1(q_1) = 275*300-200*300=22500$

Profit level for Firm 2: $p(q)q_2 - C_2(q_2) = 275*150-200*150=11250$

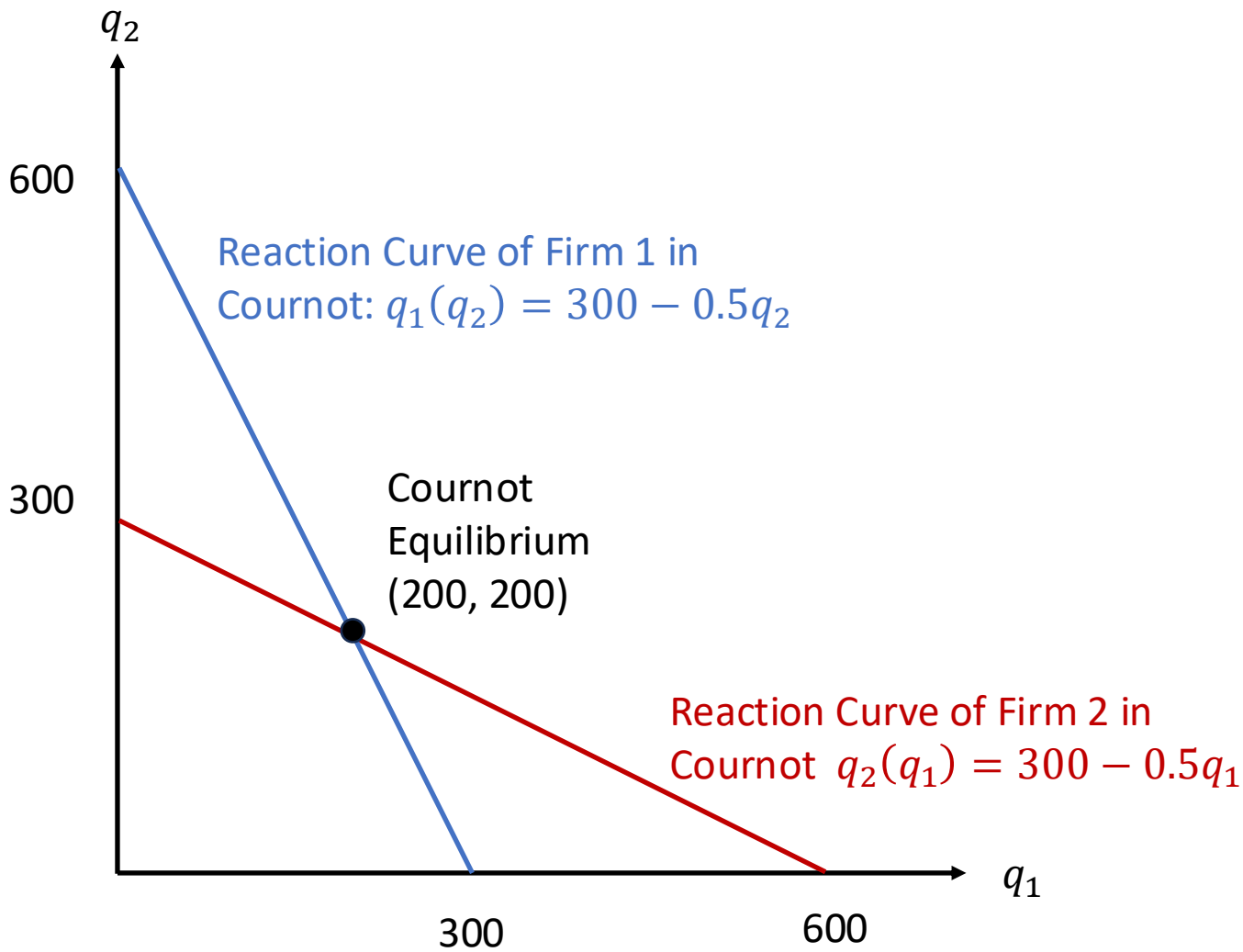
There is a first-mover advantage!

Why is there a First-Mover Advantage in Stackelberg game?

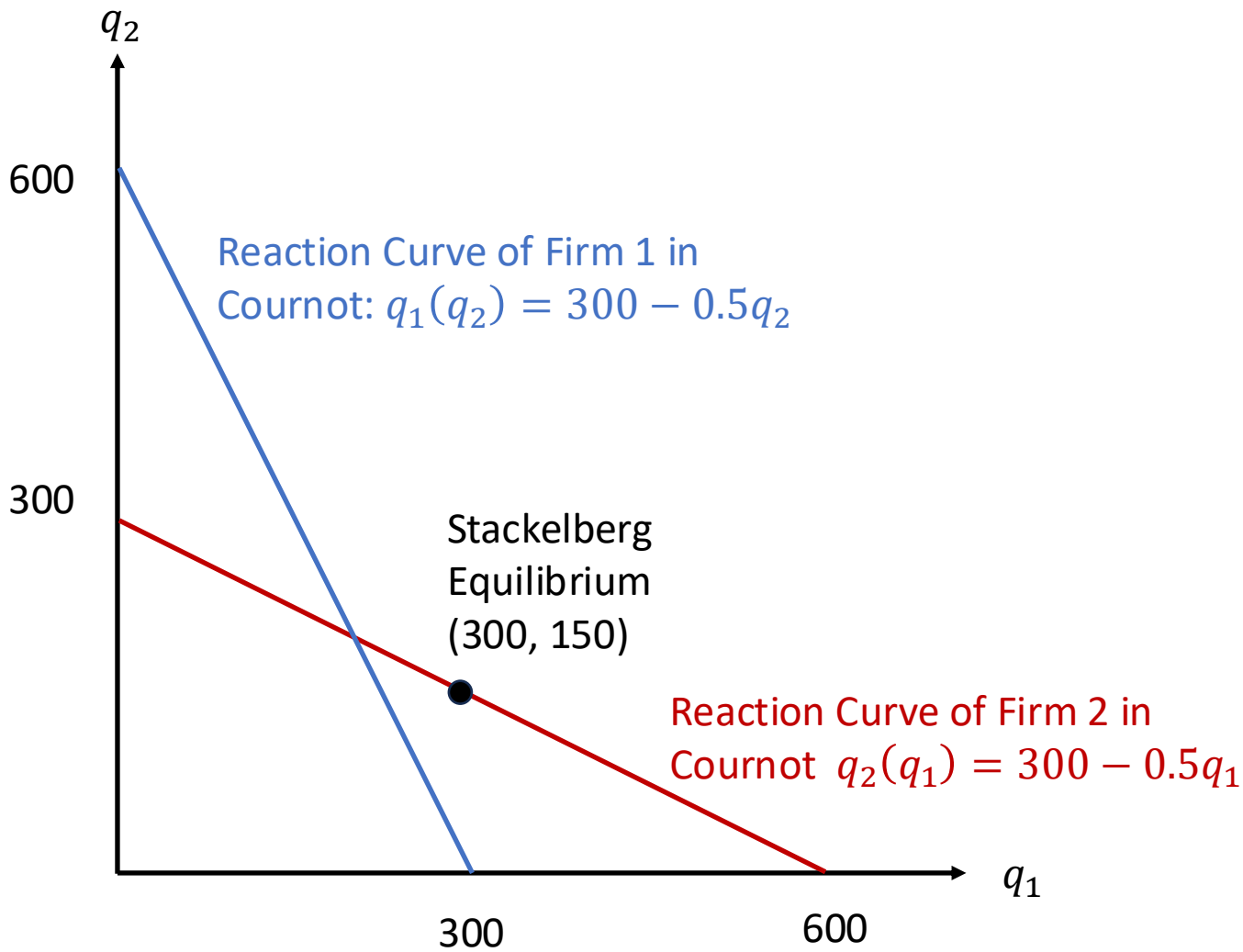
- The leader **commits** to an action that the follower must take as given and plays a best response to, so the leader **controls the follower's behavior indirectly**.

- In a Stackelberg game, producing more early makes the follower back off, so expanding output is a way for the leader to discourage competition.

Cournot Equilibrium in the Graph



Stackelberg Equilibrium in the Graph



Does the leader always have an advantage?

Not really.

Consider the same problem except that the leader (Firm 1) has a higher marginal cost: $Q(p)=1000-2p$. Leader $C_1(q_1)=250q_1$. Follower $C_2(q_2)=200q_2$.

Competing Forces:

Leader's advantage: being the first mover.

Leader's disadvantage: cost disadvantage.

Does the leader always have an advantage?

The follower's problem is identical: $q_2(q_1) = 300 - 0.5q_1$

The leader's problem:

Step 1(1): Derive the inverse demand function

- Total output $q = q_1 + q_2(q_1) = q_1 + (300 - 0.5q_1) = 300 + 0.5q_1$,
- $p(q) = 500 - 0.5q = 500 - 0.5(300 + 0.5q_1) = 350 - 0.25q_1$

Step 1(2): $\boldsymbol{R_1}$

- $R_1 = p(q)q_1 = (350 - 0.25q_1)q_1 = 350q_1 - 0.25q_1^2$

Step 1(3) : Compute $\mathbf{MR_1}$ (take derivative of $\mathbf{R_1}$ respect to q_1), $\mathbf{MC_1}$

- $\mathrm{MR}_1 = 350 - 0.5q_1$. $\mathrm{MC}_1 = 250$.

Step 2: Set $\boldsymbol{MR_1 = MC_1}$

- $350 - 0.5q_1 = 250$, isolate q_1: $q_1 = 200$ **This is Firm 1's output level.**
- $q_2(q_1) = 300 - 0.5q_1 = 200$ **This is Firm 2's output level.**

Does the leader always have an advantage: Summary

$Q(p)=1000-2p$. Leader $C_1(q_1)=250q_1$. Follower $C_2(q_2)=200q_2$.

In this example, the two firms produce identical output level, meaning that the cost disadvantage of the leader here exactly wipes out the first-mover advantage.

You can easily verify that Firm 1 earns a lower profit here.

You can check that when MC is above 250, then Firm 1 produces less than Firm 2 in Stackelberg equilibrium.

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